
pixartek
ROBOTIC PAINTING LEARNING SYSTEM

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TEAM PRESENTATION

We are PIXART TEAM, two young people passionate about technological innovation, art, and robotics, from Santiago de los Caballeros, Dominican Republic. We participate in the World Robot Olympiad with the goal of demonstrating that technology can not only solve industrial problems but also address educational and creative challenges.

Throughout the development of our project, each of us contributes our talents and skills, taking on responsibilities according to our strengths and working as a team to turn our ideas into reality. Would you like to learn more about the members of PIXART TEAM and the roles each one plays? Open the envelope and find out!

Annette Marie Fátima Vásquez González:
She is 17 years old and is responsible for the artificial vision system powered by artificial intelligence, which analyzes the painting in real time and sends feedback to the application. She also works on the development of the application, which serves as the control center for managing the user experience.

Sarah Elisa Gil Durán:
She is 17 years old and is responsible for the automatic brush-cleaning mechanical system. She also develops the mechanical system responsible for dispensing paint through pumps to begin the painting process. Additionally, she worked on the research and design aspects.

PROJECT IDEA SUMMARY

Pixartek was created in response to the difficulties many students face when learning to paint due to traditional teaching methods, the lack of personalized guidance, and the perception that art is difficult to master. We chose this problem after observing in our school environment how these barriers can limit students' interest, confidence, and artistic skill development.

Pixartek is an intelligent robotic assistant designed to teach painting in an interactive, practical, and personalized way. It uses artificial intelligence-powered cameras to observe the painter's movements and analyze brushstrokes, direction, speed, contour tracking, and color use in real time, providing instructions to correct and improve their technique. It also features an automatic paint dispensing system that prepares and provides the exact colors in the quantities needed for each stage of the artwork.

Unlike other systems, Pixartek does not paint for the student or replace the teacher. Instead, it guides students while they create their own artwork, turning artistic practice into an active learning experience. Its implementation in schools, workshops, and educational spaces would enrich classes, provide individualized guidance, and help more people develop their artistic skills, regardless of their experience.

Pixartek aims to demonstrate that robotics and artificial intelligence can strengthen art education, making learning to paint more accessible, interactive, and engaging, while encouraging human creativity and keeping the traditional practice of painting alive.

ROBOTIC SOLUTION

Pixartek is an intelligent and interactive system that combines visual projection, artificial intelligence, and robotic assistance to teach painting in real time on a canvas. Its main difference is that it does not aim to paint for the user, but rather to teach them while they create their own artwork, guiding them throughout the process and helping them achieve the necessary shades and results. In this way, Pixartek turns technology into an artistic assistant, keeping creativity and decisions in the hands of the person.

PROJECT IDEA EVOLUTION

The Pixartek project has evolved as we conducted tests and encountered difficulties that led us to develop new solutions.

Initially, we placed the projector in front of the canvas to display the image and outlines that would guide the student. However, during testing, we discovered that the person's hand and body blocked the projection while painting. For this reason, we decided to place the projector behind the canvas.

This change presented a new problem: one of the easel's components obstructed part of the projection. To solve this, we modified the easel's design, allowing the image to be projected clearly onto the canvas without any obstacles.

Initially, Pixartek had an artificial intelligence-powered camera that observed the canvas, allowing it to identify the painting's progress and provide real-time corrections to the student. Later, we decided to add a second camera with infrared technology, dedicated to tracking the paintbrush. This new camera can observe the brush in real time and analyze important aspects of the technique, such as its orientation and brushstroke speed. This way, the system not only analyzes what the student paints, but also how they paint it.

We also noticed that having to stop constantly to wash the brushes when changing colors interrupted the learning process. Therefore, we added an automatic brush-cleaning system to the robot, making the experience more comfortable and continuous.

Finally, the first versions used a Raspberry Pi 5 as the main processing unit. As the vision and artificial intelligence features increased, we decided to replace it with a Mac mini M4, which has greater processing power to handle the system's requirements.

SIMILAR IDEAS

What makes this project better than conventional methods? Most current approaches to art education, such as traditional classroom teaching, online tutorials, or passive content consumption through videos and social media, lack personalization, practical guidance, and real-time feedback. Students often learn in groups, receive little individual attention, or simply watch without actively practicing, leading to frustration, slow progress, and eventual loss of interest.

The application will include a personalized guidance and recommendations section, where artificial intelligence analyzes the user's performance and provides specific suggestions for improvement. These may include real-time corrections, technical tips, recommended exercises, and adjustments to the use of tools or paintbrushes.

Aspect	Current solutions	PIXART Plan	PIXART's main advantage
Lack of personalized feedback	Group classes with little individual attention; students repeat errors without specific correction.	AI that analyzes progress in real-time and offers personalized step-by-step guidance.	Adaptive learning that accelerates progress and reduces frustration.
Abandonment of artistic learning	Frustration from lack of guidance, low results or perception of 'having no talent'.	A system that simplifies complex techniques into clear and achievable steps, increasing motivation.	Makes art accessible for beginners, increasing retention o retention and confiance.
Disconnection between art and modern technology	Educatioal artistic traditiona separated from technological innovation.	Integration of robotics, AI and art into a single interactive educational system.	Positions art ass relevant in the technological era, attracting news generations.
Long-term cultural impact	Progressive decrease of artists, techniques and cultural transmission.	A scalable plataform that can be implemented in schools, museums and homes.	Revitalize the culture artistice and ensures its transmission to future generations.

Aspect	Current solutions	PIXART Plan	PIXART's main advantage
Decline of art education and cultural participation	Program school programs, limited., reduction of art classes, unattractive traditional teaching; unequal access according to socioeconomic level.	Plataforma interactive with step-by-step guidance through projection and AI that makes artistic learning accessible, attractive and practical for anyone.	Democratizes access to art, increasing participation even among young people without prior training and reducing the educational gap.
Loss of cultural heritoge and artistic knowledge	Theoretical teaching, little focus on practice; limited access to museoms o artistic tranning artistics; learning passing over art.	System that teaches techniques, styles and context historical while the user paints in real-time, integrating practical + cultural.	Actively preserves the art by turning the user into a creator, not just a spectator, keeping historical techniques and styles alive.
Inequality in cultural access	Greater access for people with more resources, materials artistics, classes and work ers usually costly.	Use of accessible technology (e.g., Raspberry Pi + AI) to create a low-cost solutiomn replicable in schools and communities.	Reduces economic barriers and brings quality artistics qualition to mais social sectors.
Displacement of manual painting by digital	Passive consumption of contents (social networks, videos); materials, classes and work ers usually costly.	Experience hybride; uses technology to guide, but keeps the real physical painting (brush, canvas, texture).	Reclaims motor and sensory skills while using technology as a support, not a replacement.
Loss of tactile skills and emotional connection	Lack of physical interaction; digital experiences do not develop manual skills nor dood connection.	Direct interaction with artistic materials guided by AI, fostering leaning active and sensorials.	Strengthens the emotional connection with art and develops fine motor skills.

CONSTRUCTION OF THE SOLUTION

System Components	Specific Materials
1. Intelligent Projection System	Canvas, Easel, Fixed Structure to Maintain Projection Accuracy, Caydo P1 Projector, Power Supply, Mac mini M4
2. Computer Vision System	Logitech C920 Camera, Arducam Owl Sight Camera
3. Painting System	Peristaltic Pumps, IV Tubing, Paint, Linear Rail Actuator, L298N Motor Driver, Raspberry Pi 5, Union Adapter, Luer-to-Barb Adapter, Paint Palette, Limit Switch, A4988 Driver Module, Male PVC Adapter, Fireman-Style Hose Nozzle
4. Brush Cleaning System	Submersible Pump, Infrared Proximity Sensor, PVC Pipe, PVC Elbow, Rain Irrigation Tape Adapter, Drip Hose Adapter, Hose Elbow, Flexible Shower Hose, Container
5. Application Control Panel	SAT Touchscreen, HDMI Cable, USB-to-USB Cable, Mac mini M4, Power Supplies
6. Electronic Paintbrushes	Electronic Paintbrushes with Orientation Technology and Infrared Sensors powered by AA Batteries.
7. Other Materials and Tools	Server Rack, Paintbrushes, Stickers, Spray Paint, White Acrylic Paint, Screws, Nuts, Tank Sealant, Epoxy Steel, Screwdrivers, Bucket, Grip Pliers, Soldering Iron, Cutting Pliers, Drill and Drill Bits, Hunter Clear Silicone, VEX V5 Kit, Washers, Jumper Wires, Power Supplies, Wheels, Teflon Tape Roll, Hacksaw, Metal Sandpaper, Regular Sandpaper, Electrical Insulating Tape, Circular Saw, Cables.

1. Intelligent Projection System:

The projection system was built by placing the canvas on a stable easel and mounting the Caydo P1 projector on a fixed structure. This keeps the image aligned and prevents it from moving or becoming distorted. The projector displays guides directly onto the canvas, allowing the user to follow the lines and shapes in real time.

2. Computer Vision System:

The computer vision system was developed using a Mac mini M4 connected to a camera via USB, which captures images quickly. The camera is placed on an adjustable mount to adapt to the size and angle of the canvas. An infrared camera was also installed to track the paintbrush and detect the direction and speed of its movements. The entire system is powered by a stable power supply, allowing it to monitor the user's progress in real time and generate recommendations throughout the artistic process.

3. Painting System:

Five containers hold the base paints (blue, yellow, white, black, and red). Each one is connected through IV tubing to its own peristaltic pump, which squeezes the tube with rollers to push out a precise amount of paint. The pumps are driven by L298N motor drivers, controlled by a Raspberry Pi 5. The tubes end over a paint palette with separate wells, and a linear rail actuator moves the outlets along the palette so each mixture lands in its own well. A stepper motor drives the rail through an A4988 driver, and two limit switches mark the ends of its travel. To make a color, the system doses the right amount from each pump into one well, then moves to the next well until all the colors of the artwork are ready.

4. Brush Cleaning System

The cleaning station is a container with an infrared barrier: a transmitter on one side and a receiver on the other. When the brush is placed in the container, it blocks the beam, and the system detects it. A submersible pump sits in the same container and pushes water through PVC pipes aimed at the brush. The water rinses the paint out and returns to the container, where the pump takes it again, so the same water is recycled. After multiple uses, the water gets dirty with paint, and it can simply be replaced with clean water. When the brush is removed, the beam is restored and the pump stops

5. Application Control Panel:

PIXARTEK features an on-screen application that serves as a control panel to record, organize, and analyze the creative process in real time ran by Mac Mini M4 and displayed on an SAT Touchscreen. It allows users to view their completed artworks, techniques used, practice sessions, and artistic progress. It will also record their activity to evaluate improvements in areas such as proportions, color, precision, and technique. Through artificial intelligence, it will provide personalized recommendations, real-time corrections, technical tips, exercises, and suggestions on the use of tools and paintbrushes.

Within the control panel, users will have access to:

- Progress Tracking and Personalized Tips
- Real-Time Corrections During Painting
- Step-by-Step Guidance Adapted to Their Skill Level
- Automatic Digital Portfolio with All

6. Electronic Paintbrushes

To improve analysis accuracy, the robot features paintbrushes that use positioning technology and infrared sensors to determine the direction, speed, and alignment of each brushstroke.To build each electronic paintbrush, we took a flashlight and replaced its light with an infrared LED. Then we 3D-printed an adapter that holds a regular brush head on one side and the flashlight on the other. The result is a brush that looks normal but gives off a light that only the infrared camera can see.

SCAN THE QR CODE

After scanning, you will have access to a collection of videos and images showcasing the entire construction process of Pixartek.

PIXARTEK JOURNAL

PIXARTEK VIDEO

PROGRAMMING OF THE SOLUTION

Projection System

The projector is controlled directly by the Mac's SwiftUI application. The app detects the projector screen, connected via HDMI, and opens a full-screen black window on it. When a stage begins, it displays the outline for that stage and projects it onto the canvas. When the session ends, the projector returns to a black screen.

Vision System

The vision system uses OpenCV, a library for image analysis. A color camera (C920) monitors the canvas, while an infrared camera (Arducam Owl Sight) tracks the paintbrush. Both are connected to the Mac Mini M4. A Python service, launched by the application itself and listening on the local port, receives the camera frames. Using OpenCV, it converts them from BGR to Lab and HSV, color spaces that separate color tone from brightness, allowing colors to be compared even when lighting conditions change. Using a color mask, it measures how much of the canvas has been painted and compares the painted areas with the stage image, detecting contours and measuring which parts of the painted area remain inside or outside the boundaries. For the infrared camera, the same service applies an adaptive threshold to detect the paintbrush's LED and track its position frame by frame. On top of this image analysis, five bots work in parallel, each one running as an independent process. None of them follows a script; each one makes decisions in real time based on what it sees and its trained knowledge. The brush bot combines OpenCV with an RF-DETR detector trained to recognize which paintbrush is being used and whether it is the correct one for that area. The color bot compares the colors on the canvas with the artwork's palette, using Lab color ranges to verify them. The contour bot compares the canvas with the stage image and determines whether the user is painting outside the lines or in the wrong area. The trajectory bot calculates the paintbrush's speed and direction using the infrared camera. The scene bot uses the CLIP model to classify what is happening on the canvas: a hand present, painting, cleaning, or nobody.The bots only observe and detect; each one sends what it detects as JSON messages to Hermes. Hermes is a Python service that interprets the errors reported by the bots, turns them into friendly suggestions and corrections, and says them out loud to the user in real time.

Painting System

The painting system runs on the Raspberry Pi 5, and the Mac mini M4 sends it requests through HTTP. There are five peristaltic pumps, one for each base color (blue, yellow, white, black, and red). Each pump is connected to a driver and controlled through two pins, IN1 and IN2, using the lgpio library on the Pi 5. During testing, we noticed that each pump dispenses a different amount of paint per second, so we measured the flow rate of each one and stored it in the code in milliliters per second. To dispense a specific amount, the program divides the required milliliters by the pump's flow rate, which determines how many seconds it needs to remain on. To prepare an artwork, the service receives the colors for each stage and calculates the recipe for each one using the primary colors, comparing the result with the Pantone color selected for that shade. The stepper motor that moves the paint container over the palette is controlled by a Raspberry Pi 4 with an A4988 driver. The Pi 5 sends it commands through a serial cable (UART at 9600 baud, with a common ground). The motor always runs at the same speed, 3,200 pulses per second. Two limit switches, MIN and MAX, stop the carriage as soon as they are triggered. The zero position is set when the carriage reaches the MIN limit switch. From that point, the program knows how many steps away each of the five color wells is, allowing it to move directly to any of them.

Brush Cleaning System

An infrared barrier detects when the paintbrush reaches the cleaning area. It consists of an infrared transmitter and a 38 kHz receiver. Since the receiver becomes inactive when exposed to continuous light, the transmitter emits 80 ms bursts every 300 ms, generated using hardware PWM. This timing also serves as the heartbeat for the service, which runs continuously on the Pi 5. When several consecutive bursts confirm that something is interrupting the beam, preventing the system from reacting to a single reading, the submersible water pump turns on for 5 seconds. During this time, all other inputs are ignored. Once this period ends, the system checks again: if the paintbrush is still there, it runs the pump for another 5 seconds; if not, it resumes monitoring.

Central Server and Mac Mini M4 Control Panel:

The Mac mini M4 is the center of PIXARTEK: all the software runs on it, and it sends commands to the Raspberry Pis. Its operation can be divided into three parts: the application used by the student, the background services, and the way everything communicates. The application is programmed in SwiftUI and works as the control panel. In it, the student's name is entered, an artwork is selected from the catalog, progress through each stage is monitored using the camera and paintbrush, and the user moves from one stage to the next. It opens in full-screen kiosk mode and uses two displays, one for the controls and the other for the projector. When a session begins, the app starts the vision service, turns on the cameras, and informs Hermes of the selected artwork and stage. When the session ends, it shuts everything down and leaves the projector screen black. Behind the application, Python services run independently, each as a separate process with its own port. These include the vision service (8765), the five bots (brushes 8766, color 8772, trajectory 8773, contour 8774, and scene 8775), Hermes (8770), and Chatterbox voice (8771). Ollama also runs the hermes3:8b language model, along with Qdrant, the database where Hermes stores its knowledge. They are managed by launchd, the macOS startup system, which launches them automatically when the Mac turns on and allows one to be restarted without affecting the others. Everything communicates through HTTP using JSON messages. The bots send Hermes what they detect, the app sends Hermes the session status, and the app requests paint quantities and rail movements from the Pi 5. This allows the Mac to decide what to do while the Raspberry Pis execute the commands.

Electronic Paint Brushes

The Arducam Owl Sight infrared camera, fixed above the canvas, sees the brush's light as a bright point. A Python service on the Mac finds it in every frame with an adaptive threshold. Speed is measured in centimeters per second: slow under 2 cm/s and fast over 13 cm/s, or 10 cm/s in thin areas. Direction comes from the main axis of the whole stroke and it only corrects when the stroke deviates from the recommended direction.

DEVELOPMENT CHALLENGES

One of the main challenges was defining how real-time feedback should work. Initially, we considered a system similar to traditional projected tutorials, where users simply follow visual instructions. However, this did not solve the main problem: the lack of personalized corrections. We realized that without real-time analysis, users could repeatedly make mistakes without noticing them. This led to a key decision: integrating computer vision capable of interpreting the user's brushstrokes and responding dynamically.

Another important challenge was achieving accurate alignment between the projection and the canvas. Even small errors could affect the learning experience. This required developing advanced calibration systems to ensure that the visual guide matched the physical surface precisely.

We also faced the challenge of balancing technology with artistic experience. A system that was too automated could make users completely dependent on the machine, reducing their creativity. For this reason, Pixartek was designed as a support tool, not a replacement, keeping the person as the main creator.

However, one of our biggest challenges was color mixing. Initially, the robot could not mix colors correctly. For example, when we wanted blue, the robot would dispense a color that looked more like gray. We had to work extensively to help the robot correctly interpret the combinations needed to create each specific color.

Finally, integrating real-time artificial intelligence represented a significant technical challenge. Processing images, interpreting brushstrokes, and responding instantly requires edge computing optimization. This decision was essential to ensure that the system operates smoothly without relying entirely on the cloud.

SOCIAL IMPACT AND INNOVATION

Did you know that the world is experiencing a silent crisis in art education and the preservation of traditional cultural practices? In many countries, opportunities to learn painting, understand the history behind artists, and practice hands-on techniques are decreasing. This loss affects not only individual skills but also cultural identity and the way societies express themselves and connect with one another. Pixartek was created to address this issue. It preserves the history of artistic styles and artists, brings traditional painting back to life, and provides personalized guidance so that everyone can learn and develop at their own pace. In our community, it will mainly benefit students and schools looking for new ways to learn art, especially those who do not have access to personalized art education.

The Decline of Art Education and the Loss of Cultural Participation:

Eurostat statistics, published in the Culture Statistics – Cultural Participation (2026) report, show that cultural participation decreased between 2015 and 2022 in almost every European Union country. Furthermore, participation is linked to socioeconomic status: people with lower incomes and educational levels participate less in cultural activities than those with greater resources. These differences reflect inequalities in access to culture.

In England, the figures are particularly concerning. According to the Cultural Learning Alliance Report Card 2025, enrollment in GCSE Arts subjects has decreased by 42% since 2010, while the number of Arts teachers has fallen by 27%. This decline limits opportunities to learn and pass artistic techniques on to new generations.

This is not just an educational trend; it is a cultural loss. When fewer people participate in art, fewer stories behind artists are preserved, fewer traditional techniques are taught, and the risk of traditional practices disappearing increases.

Loss of Cultural Heritage and Artistic Knowledge:

As participation in the arts decreases, exposure to the history, techniques, and stories of influential artists also declines. This phenomenon is not isolated; it reflects a global trend documented by multiple studies on cultural participation.

According to Eurostat data and European cultural studies, people with lower educational and economic backgrounds participate significantly less in cultural activities, creating a direct gap in access to artistic heritage. This inequality not only limits learning but also determines who has the opportunity to discover, practice, and preserve art. In many cases, access to museums, art education, and cultural experiences can be up to twice as high among people with higher educational levels.

Furthermore, international reports on art education indicate that, in several countries, the reduction of school art programs has caused entire generations to grow up with little or no exposure to artistic movements, traditional techniques, and key historical figures. This means that artistic knowledge is no longer actively passed down and instead becomes something distant, theoretical, or even nonexistent for many young people.

The result is concerning: historical artistic styles, movements, and stories are no longer actively practiced or taught, increasing the risk of their disappearance over time. It is not just about losing techniques, but also ways of seeing the world, cultural identities, and forms of expressing the human experience.

The Shift from Traditional Painting to Digital Activities and the Loss of Tactile Skills:

Participation in traditional forms of art is declining not only because of a lack of formal education, but also because many people, especially young people, consume art and culture through digital media rather than practicing hands-on techniques. The studies Arts Participation in 2022: A Technical Summary Report, by the National Endowment for the Arts, and New Opportunities for Interest-Driven Arts Learning in a Digital Age, by The Wallace Foundation, show how people participate in artistic activities both in person and through digital media. This highlights the importance of providing opportunities to practice creative activities that require physical interaction with artistic materials.

Fine motor and sensory skills developed through physical interaction with paintbrushes, textures, and surfaces may receive less practice. The emotional connection to the creative process may also be affected, since the tactile experience of art cannot be fully replicated on screens. Additionally, cultural narratives may become more fragmented as opportunities to learn artistic techniques through practice decrease. Creating an artwork allows people to understand aspects of the process that are not always experienced through digital observation alone.

Lack of Personalized Feedback in Art Education:

The study The Power of Feedback Revisited: A Meta-Analysis of Educational Feedback Research, by Wisniewski, Zierer, and Hattie (2020), demonstrates that feedback has a positive influence on learning. Meanwhile, Artificial Intelligence-Supported Art Education (2022) examines the use of artificial intelligence to improve art education. The lack of personalized guidance can make it difficult to correct mistakes and develop skills, making specific feedback important for improving artistic learning.

IMPACT OF THE SOLUTION ON SOCIETY

The Pixartek project is designed to address the challenges of traditional art education through an innovative solution that revives traditional painting, preserves artistic heritage, and offers a personalized and interactive learning experience, generating a cultural and educational impact. Various recent studies support this approach, demonstrating that artificial intelligence can significantly transform art education through personalized feedback and real-time analysis. For example, a study published in Scientific Reports (2023) showed that these systems can analyze artworks and provide recommendations tailored to the user's skill level, improving abilities such as proportion, color, and composition.

Likewise, more recent studies (PMC, 2025) highlight that implementing artificial intelligence in painting courses increases motivation, confidence, and learning retention, while a review published in MDPI (2024) concludes that these technologies enhance creativity, active participation, and understanding of cultural contexts. Along the same lines, research published in SAGE Journals (2025) demonstrates that real-time feedback allows for continuous monitoring of student progress. However, these studies also point out that most current solutions are limited to digital environments and do not effectively integrate physical artistic practice, leaving a clear opportunity for innovation. In this context, Pixartek emerges as a solution that combines artificial intelligence, computer vision, and technological assistance in a physical system, bridging this gap.

PIXARTEK's Beneficiaries and Educational and Cultural Impact:

Pixartek will help people who face limitations in accessing structured art instruction or personalized feedback. Many learners struggle to make progress in painting due to a lack of real-time guidance. With Pixartek, users receive step-by-step assistance directly on the canvas, allowing them to learn at their own pace while improving their accuracy, technique, and confidence. Additionally, Pixartek reconnects users with the emotional and cultural value of art by integrating the stories and styles of historical artists into the learning process. This transforms painting into an immersive experience where users not only create art but also understand its origins, context, and evolution. In this way, users develop both technical skills and a deeper appreciation for artistic heritage.

Cultural Preservation and Artistic Heritage:

Pixartek directly contributes to the preservation of artists and styles that are being forgotten by integrating them into the learning process. Instead of art history being purely theoretical, it becomes a practical and interactive experience. Users can explore different artistic movements, recreate techniques, and connect with styles that could disappear due to a lack of teaching or exposure. Additionally, Pixartek incorporates generative AI capable of interactively recreating the faces, voices, and attitudes of historical artists, allowing them to "come to life" within the experience. Through this interaction, the artists themselves can explain their processes, techniques, and approaches, making learning more immersive, emotional, and meaningful. This helps keep artistic traditions alive through active practice and a deeper connection with those who created them.

Educational Accessibility and Personalized Learning:

Pixartek works as a personalized teaching assistant that adapts to each user's pace and skill level. Various reviews, such as the PRISMA review, show that adaptive learning improved academic performance in 59% of the cases analyzed. Pixartek applies this principle by providing real-time guidance, corrections, and suggestions throughout the painting process.

This reduces frustration, accelerates progress, and allows each person to gradually develop their skills without constantly depending on a human instructor.

Revitalization of Hands-On Artistic Practice:

Instead of replacing traditional painting with digital alternatives, Pixartek promotes the use of physical canvases and real materials. This is important because hands-on practice develops fine motor skills, spatial perception, and a deeper sensory connection with the creative process. Through projection and intelligent assistance, Pixartek guides users directly on the canvas, encouraging the continuation of traditional techniques in a modern context.

Social Impact and Emotional Well-Being:

Beyond technical learning, Pixartek also contributes to emotional well-being and personal expression. Various studies have shown that participating in artistic activities can reduce stress, improve mental health, and increase a sense of achievement. By making painting more accessible, guided, and less intimidating, Pixartek allows more people to experience these benefits, encouraging creativity, confidence, and individual expression.

Pixartek is not just an educational tool; it is an innovative social and cultural initiative that:

- Preserves the stories of artists and styles that are being forgotten.
- Revives the hands-on practice of painting in a digitalized world.
- Offers personalized feedback that transforms the art-learning experience.
- Contributes to emotional well-being and social cohesion.

These impacts are supported by real data showing the decline in artistic participation, the reduction of formal art education, and the need for deeper and more meaningful creative experiences.

POSSIBLE NEGATIVE EFFECTS

One possible negative effect is that people who learn to paint with Pixartek may later develop a dependency on the system, limiting their ability to create their own style. This potential negative effect only applies to users who are interested in learning to paint in their own way, without Pixartek. However, Pixartek will always offer options to practice painting while gradually reducing its assistance, encouraging users to progress independently and develop their own style.

PRACTICAL USE CASES

Pixartek has multiple practical use cases in different settings, especially in education, culture, and independent learning. In schools and art academies, it allows students to learn painting techniques through guided instruction and real-time feedback, supporting the development of skills from basic to advanced levels. It can also be used in museums and cultural spaces, allowing visitors to interact with artworks, understand their creative process and historical value, and recreate them. Another use for Pixartek is as a form of entertainment for elderly people living in nursing homes, allowing them to create paintings that they can give as gifts or sell to raise funds. Additionally, it contributes to the preservation of art, history, and culture by making it possible to digitize and reproduce traditional styles. On a personal level, anyone can use Pixartek at home as an accessible tool to learn and practice painting independently, making art a more engaging, modern, and interactive experience.

USE OF AI

GROK:

We used Grok to support technical problem-solving and quickly find information related to hardware, components, and system operation. It also helped with image generation.

CLAUDE:

Claude was used primarily through programming to develop and organize parts of the project's code. It helped us structure the system's logic, create automations, integrate components, and solve errors during technical development.

CHAT-GPT:

It was used to organize ideas, develop the overall project structure, and prepare presentations. It also helped with research related to hardware and the development of the narrative used to communicate PIXARTEK's purpose and operation.

QWEN:

It was used for in-depth research on issues related to art, art education, and cultural preservation. It also helped analyze the official WRO document, understand the mission areas, and generate supporting images for visual concepts and project references.

KLING AI:

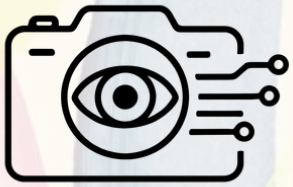
It was used to create videos and visual animations for the project. It allowed us to transform ideas, images, and scenes into more dynamic audiovisual content to explain Pixartek.

LIST OF RESOURCES

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Projection



Vision



Assistance

¿What if we could learn to paint with real-time robotic guidance? That is the question we asked ourselves, and it is the idea behind PIXARTEK. Instead of waiting for frustration to turn into giving up, we created a system that addresses the problem before it happens.

Here's how it works: We install an intelligent projection system and computer vision cameras in creative spaces such as workshops or schools. These are not ordinary tools; they use artificial intelligence to analyze what the user is painting in real time, detecting errors in brushstrokes, colors, or proportions.

As soon as they detect something, they send immediate feedback to the autonomous robot. The robot does not wait for commands; it moves independently, prepares the exact colors using dispensing pumps, cleans the paintbrushes automatically, and guides the painting process with precision.

PIXARTEK is more than a robot; it is a form of intelligent art preservation. It is our solution to keeping people connected to their creativity, preventing the loss of artistic skills, and protecting our cultural heritage at the same time. With this project, we demonstrate that technology can be used to solve real-world problems and make the world a more creative and accessible place.

PIXART TEAM